

Problem

(a) Obtain the voltage V_o in the circuit of Fig. 2.127(a).

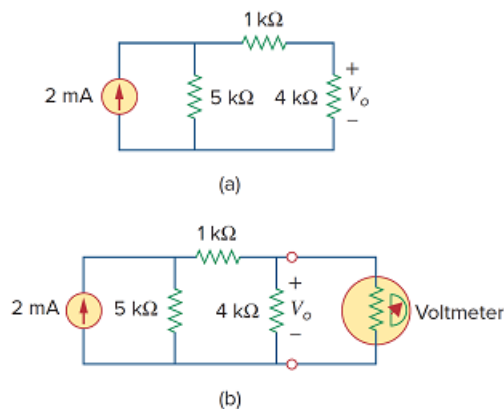
(b) Determine the voltage V'_o measured when a voltmeter with 6-k Ω internal resistance is connected as shown in Fig. 2.127(b).

(c) The finite resistance of the meter introduces an error into the measurement. Calculate the percent error as

$$\left| \frac{V_o - V'_o}{V_o} \right| \times 100\%$$

(d) Find the percent error if the internal resistance were 36 k Ω .

Figure 2.127



Step-by-step solution

Step 1 of 7

(a)

Consider the following circuit diagram:

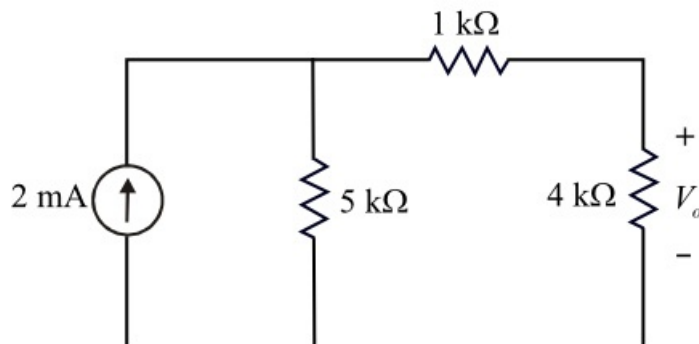


Figure 1

Step 2 of 7

Apply current division rule, to find the current through $4\text{ k}\Omega$ resistor.

$$\begin{aligned} I_o &= (2 \times 10^{-3}) \left(\frac{5\text{k}}{5\text{k} + 1\text{k} + 4\text{k}} \right) \\ &= (2 \times 10^{-3}) \left(\frac{5\text{k}}{10\text{k}} \right) \\ &= 1\text{ mA} \end{aligned}$$

Calculate the voltage across the $4\text{ k}\Omega$ resistor using Ohm's law.

$$\begin{aligned} V_o &= (4\text{k})(I_o) \\ &= (4\text{k})(1 \times 10^{-3}) \\ &= 4\text{ V} \end{aligned}$$

Therefore, the voltage across the $4\text{ k}\Omega$ resistor, V_o is 4 V.

Step 3 of 7

(b)

Consider the following circuit diagram:

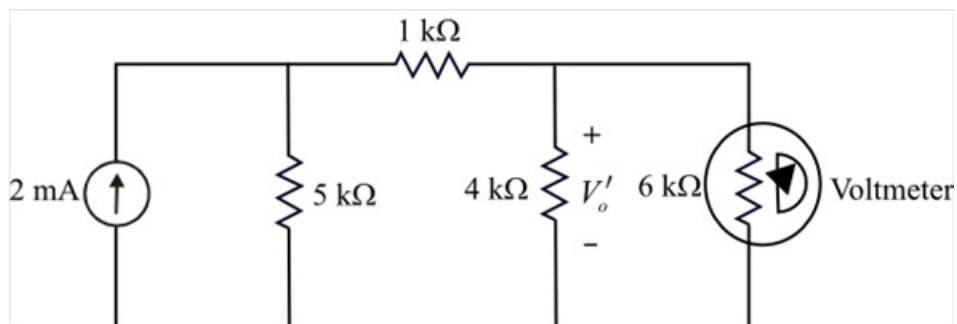


Figure 2

Step 4 of 7

In the figure 2, the $4\text{ k}\Omega$ and $6\text{ k}\Omega$ resistors are connected in parallel. Calculate the resultant equivalent resistance.

$$\begin{aligned} R_{eq} &= 4\text{k} \parallel 6\text{k} \\ &= \frac{(4\text{k})(6\text{k})}{4\text{k} + 6\text{k}} \\ &= 2.4\text{ k}\Omega \end{aligned}$$

Consider the following reduced circuit of Figure 2.

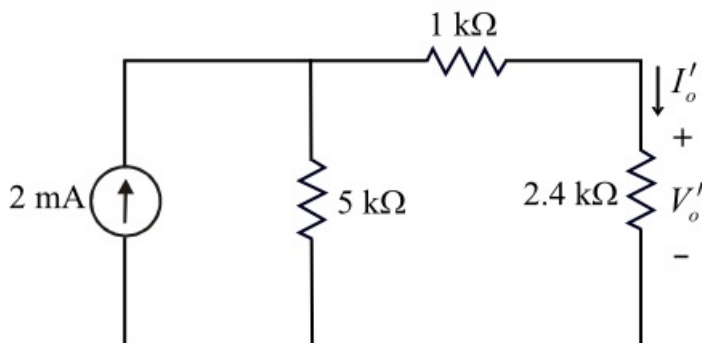


Figure 3

Step 5 of 7

Apply current division rule, to find the current through $2.4 \text{ k}\Omega$ resistor.

$$\begin{aligned} I'_o &= (2 \times 10^{-3}) \left(\frac{5\text{k}}{5\text{k} + 1\text{k} + 2.4\text{k}} \right) \\ &= (2 \times 10^{-3}) \left(\frac{5\text{k}}{8.4\text{k}} \right) \\ &= (2 \times 10^{-3}) (0.595) \\ &= 1.19 \text{ mA} \end{aligned}$$

Calculate the voltage across the $2.4 \text{ k}\Omega$ resistor using Ohm's law.

$$\begin{aligned} V'_o &= (2.4\text{k})(I'_o) \\ &= (2.4\text{k})(1.19 \times 10^{-3}) \\ &= 2.856 \text{ V} \end{aligned}$$

Therefore, the voltage across the $2.4 \text{ k}\Omega$ resistor, V'_o is $\boxed{2.856 \text{ V}}$.

(c)

Calculate the percent error.

$$\left| \frac{V_o - V'_o}{V_o} \right| \times 100\% \dots\dots (1)$$

Substitute 4 V for V_o and 2.856 V for V'_o in equation (1).

$$\begin{aligned} \left| \frac{V_o - V'_o}{V_o} \right| \times 100\% &= \left| \frac{4 - 2.856}{4} \right| \times 100\% \\ &= 0.286 \times 100\% \\ &= 28.6\% \end{aligned}$$

Therefore, the percent errors is $\boxed{28.6\%}$.

Step 6 of 7

(d)

Internal resistance of voltmeter is changed to $36 \text{ k}\Omega$.

The $4 \text{ k}\Omega$ and $36 \text{ k}\Omega$ resistors are connected in parallel. Calculate the resultant equivalent resistance.

$$\begin{aligned} R_{eq} &= 4\text{k} \parallel 36\text{k} \\ &= \frac{(4\text{k})(36\text{k})}{4\text{k} + 36\text{k}} \\ &= 3.6 \text{ k}\Omega \end{aligned}$$

Apply current division rule, to find the current through $3.6 \text{ k}\Omega$ resistor.

$$\begin{aligned} I''_o &= (2 \times 10^{-3}) \left(\frac{5\text{k}}{5\text{k} + 1\text{k} + 3.6\text{k}} \right) \\ &= (2 \times 10^{-3}) \left(\frac{5\text{k}}{8.4\text{k}} \right) \\ &= 1.042 \text{ mA} \end{aligned}$$

Calculate the voltage across the $3.6 \text{ k}\Omega$ resistor using Ohm's law.

$$\begin{aligned} V''_o &= (3.6\text{k})(I''_o) \\ &= (3.6\text{k})(1.042 \times 10^{-3}) \\ &= 3.75 \text{ V} \end{aligned}$$

Step 7 of 7

Calculate the percent error.

$$\left| \frac{V_o - V_o''}{V_o} \right| \times 100\% \dots\dots (2)$$

Substitute 4 V for V_o and 3.75 V for V_o'' in equation (2).

$$\begin{aligned} \left| \frac{V_o - V_o''}{V_o} \right| \times 100\% &= \left| \frac{4 - 3.75}{4} \right| \times 100\% \\ &= 0.0625 \times 100\% \\ &= 6.25\% \end{aligned}$$

Therefore, the percent errors is 6.25%.